Quadratic formula

1

a $a = 8, b = -3, c = 0$

b $a = 8, b = 0, c = 3$

c $a = 5, b = 6, c = -10$

d $a = 3, b = -17, c = 9$

2

a $x = -7, -4$

b $x = -\frac{1}{2}, -1$

c $x = 3, -\frac{5}{4}$

3 -2

4

a -9

b 0

5

a $x = \frac{5 \pm \sqrt{33}}{2}$

b $x = \frac{1 \pm \sqrt{161}}{8}$

c $x = \frac{-15 \pm \sqrt{193}}{4}$

6

a $x = 0.405, -7.405$

b $x = -0.260, -1.540$

7 $x = 0.4, -3.3$

8

a $x = \pm\sqrt{7}$

b $x = \pm 2.6$

9

a $x = 5, -1$

b $x = -4, -5$

Discriminant

10 That there is one real solution.

11 $b^2 - 4ac > 0$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12

a
i 0 ii 1

b
i -76 ii 0

c
i 1 ii 2

13

a Yes **b** No **c** Yes

14

a 0

b The roots are equal and rational.

15

a 256

b The roots are unequal and rational.

c $x = 12, -4$

Quadratics with complex solutions

16 0

17 0

18

a $x = -1 \pm 2i$ **b** $x = 3 \pm 4i$ **c** $x = -5 \pm \sqrt{3}i$ **d** $x = 2 \pm 5i$

e $x = 5 \pm \sqrt{2}i$ **f** $x = -\frac{5}{2} \pm \sqrt{2}i$ **g** $x = \frac{1}{2} \pm \sqrt{3}i$ **h** $x = \frac{2}{3} \pm \frac{2}{3}i$

i $x = -5 \pm \sqrt{15}i$ **j** $x = 2 \pm 4\sqrt{2}i$ **k** $x = -\frac{4}{3} \pm \sqrt{11}i$

19 $x = -\frac{7}{16} \pm \frac{\sqrt{47}}{16}i$

20

a Yes **b** No **c** No **d** Yes

21

a 0 **b** 0 **c** 2

22 $m > -1$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 $n = \pm 9$

24 $k = 1$

25

a $64 - 8k$

b

i $k = 8$ **ii** $k \leq 8$ **iii** $k > 8$ **iv** $k < 8$

26

a $52 - 12k$

b

i $k = \frac{13}{3}$ **ii** $k \leq \frac{13}{3}$ **iii** $k > \frac{13}{3}$ **iv** $k < \frac{13}{3}$

Discriminant and parabolas

27 None

28

a $m < -\frac{27}{4}$ **b** $m = -7$

29 $k = 16$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

30

a
i $a > 0$ ii $b^2 - 4ac > 0$

b
i $a < 0$ ii $b^2 - 4ac = 0$

c
i $a < 0$ ii $b^2 - 4ac < 0$

d
i $a > 0$ ii $b^2 - 4ac < 0$

e
i $a > 0$ ii $b^2 - 4ac < 0$

f
i $a > 0$ ii $b^2 - 4ac > 0$

g
i $a < 0$ ii $b^2 - 4ac = 0$

h
i $a < 0$ ii $b^2 - 4ac > 0$

i
i $a > 0$ ii $b^2 - 4ac = 0$

j
i $a < 0$ ii $b^2 - 4ac < 0$

k
i $a < 0$ ii $b^2 - 4ac = 0$

l
i $a < 0$ ii $b^2 - 4ac > 0$

m
i $a < 0$ ii $b^2 - 4ac < 0$

n
i $a > 0$ ii $b^2 - 4ac = 0$

o
i $a < 0$ ii $b^2 - 4ac > 0$

p
i $a > 0$ ii $b^2 - 4ac < 0$

Applications

31

a $0 < p < q$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

c $8q(q - p)$

d Two real solutions

e $x = 3p$

32

a $200 = -5t^2 + 56t + 26$

b $x = \frac{28}{5} \pm \frac{\sqrt{86}}{5}i$

c The solution is not a real number, so the cannon-ball will not reach a height of 200 meters.

33 $x = 8.7$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647